

Juliann Zhu

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EDUCATION

University of Waterloo, GPA: 3.7/4.0

BCS, Honours Computer Science with Co-operative Program

President Scholarship of Distinction

Waterloo, ON

Sep. 2025 - Present

TECHNICAL SKILLS

Languages: Python, JavaScript, HTML/CSS, C/C++, Racket, SQL, Bash, Swift

Data and Analytics: MySQL, MongoDB Atlas, InfluxDB, pandas, NumPy, Grafana

Frameworks and APIs: React, Gemini API, Node.js, REST APIs, SQLAlchemy, Uvicorn, Resend API

Tools and Systems: Git/Github, Linux, Google Cloud Platform, VirtualBox, VS Code, Xcode

RELEVANT EXPERIENCE

ORBCOMM, SKYWAVE, Network Services Department

Network Engineering Intern

May 2026 - Present

Ottawa, ON

- Developed 8 Grafana dashboards with MySQL data sources to monitor satellite network performance across 12 metrics, cutting load times by 86% through dynamic filtering and optimized SQL queries
- Shipped an AI-driven analytics service that generates summaries, detects unusual network activity, extends predictive forecasting by 60x, cuts pipeline runtime by 78%, lowers false-positive anomaly alerts by 80%
- Delivered the system's architecture and new AI-driven monitoring features to engineering leadership, backed by a high-level design document and live technical demo of visualization and usability improvements

Usable Security & Privacy Research, Women in Math DRP, University of Waterloo

Undergraduate Research Mentee

Jan. 2026 -Present

Waterloo, ON

- Co-authoring an academic research paper for the Symposium On Usable Privacy and Security investigating how users seek security and privacy advice from LLMs, with findings to be presented at the Annual Celebration of Women in Computing 2026 Conference (CAN-CWiC) and IEEE S&P
- Analyzing 2 LLMs against 78 privacy and usable security prompts sourced from 54 online databases, evaluating advice against expert feedback from 3 security researchers to assess accuracy and risks

PROJECTS

CaneOS, Hack the 6ix | Swift, Python, C++, SwiftUI, MongoDB Atlas, Auth0, Gemini API, Vercel, Websockets

Full Stack & Embedded Developer

Jul. 2026

Toronto, ON

- Built a clip-on white cane module that detects overhead hazards within a 4m range for visually impaired users, pairing ToF sensors and 30 FPS YOLOv6 camera detection on an Arduino UNO Q feed into a Python/FastAPI pipeline to classify hazards via Gemini vision and narrate within 3 seconds via ElevenLabs
- Developed companion SwiftUI iPhone and Apple Watch apps with directional haptics, 4-tier SOS alerts, and Auth0-secured sync of incidents and contacts to MongoDB Atlas via a Node.js serverless proxy

TrulyHer, Technova | React, Javascript, HTML, CSS, Base44, VS Code

AI Integration & Front-End Developer

Sep. 2025

Waterloo, ON

- Developed an AI-powered web application with speech and text input helping women in computer science manage imposter syndrome, featuring journaling and mood detection across 11 tracked emotional states
- Designed an interactive dashboard visualizing emotional trends and personalized insights from 200+ journal entries across 5 chart views, encouraging mental health awareness and empowering underrepresented groups

Project Tech Careers Hackathon | Javascript, HTML, CSS, VS Code

Platform Designer and UI/UX Developer

May - Jun. 2024

Ottawa, ON

- Designed a 4-stage platform, curating 100+ resources, to help women in computer science industries at different stages of their education find mentorship, scholarships, programs, and community resources
- Awarded the *Gender Equality Track Award* over 45 competing teams for innovation, design, and impact in addressing the UN Sustainable Development Goals by addressing the gender gap